

DESCRIPTION OF XML FILE FOR LIVER CONTOUR ANNOTATIONS

The provided “*contour.xml*” files contain the 2D image contours and their corresponding 3D model contours used for registration. These contours are classified in 3 types: *Ridge*, *Ligament* and *Silhouette*. For the *Ridge* and *Ligament*, the 2D contours are provided as (x,y) pixel coordinates, while the 3D contours are provided as vertex indices of the preoperative 3D model. For the *Silhouette*, only the 2D contours are provided. The (x,y) pixel coordinates start from (1,1) and have their origin at the top-left corner of the image. The vertex indices start from 0 and correspond to the preoperative 3D model of the patient under analysis. There is a main `<contours>` node that contains all the data, while every individual contour is defined by a `<contour>` node. For every contour, there is a `<imagePoints>` node that defines the 2D contour points with their pixel coordinates inside the `<x>` and `<y>` nodes, and also a `<modelPoints>` node that defines the corresponding model points as vertex indices inside the `<vertices>` node.

The structure of the XML file is as follows:

```
<?xml version="1.0" encoding="utf-8"?>
<contours> ── Global node of contours
  <numOfContours>3</numOfContours> ── Total number of contours
  <contour> ── Individual contour node
    <contourType>Ridge</contourType> ── Contour type (Ridge, Ligament, Silhouette)
    <imagePoints> ── Image contour node
      <numOfPoints>566</numOfPoints> ── Number of image points
      <x>859,860,861,862,863,864,865,866,867,868,869,870,871,872
      <y>1080,1079,1078,1077,1076,1075,1074,1073,1072,1071,1070,
      ── x,y point coordinates
    </imagePoints>
    <modelPoints> ── Model contour node
      <numOfPoints>9</numOfPoints> ── Number of model points
      <vertices>223,224,289,329,501,521,574,695,719</vertices> ── Vertex indices of
      ── model points
    </modelPoints>
  </contour>
  <contour> ── Individual contour node
    <contourType>Ligament</contourType>
    <imagePoints>
      <numOfPoints>331</numOfPoints>
      <x>1325,1325,1325,1325,1324,1324,1324,1324,1324,1324,
      <y>496,495,494,493,492,491,490,489,488,487,486,485,484,483
    </imagePoints>
```